# TrafficFlowBench 2026: Method and Analysis Report

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## 1. Summary

TrafficFlowBench is a synthetic freeway benchmark. It covers ten panels (corridor directions) on five corridors, with nine training months (June 2030 to February 2031) and two scored months: March 2031 for the public leaderboard and April 2031 for the private one. There are four tasks: reconstruct masked speed and flow (Task 1), keep that reconstruction physically consistent (Task 3, scored on the Task 1 answer), forecast which links are queued over the next 30 minutes (Task 2), and estimate origin–destination path flows from link counts (Task 4).

We solved the four tasks as separate problems. The composite score is a weighted sum of task scores, and Task 3 is scored only on Task 1 rows. This let us decompose the public leaderboard: each submission changed one task file against a fixed reference, and the score change divided by the task weight gave the change in that task's score (Section 2.2, Appendix A).

| | **Task 1 + 3 (state, physics)** | **Task 2 (queue)** | **Task 4 (ODME)** | **Public LB** |
|---|---|---|---|---|
| Organiser baseline | 0.35·S_state + 0.15·S_physics ≈ 0.296 | S_queue ≈ 0.302 | S_ODME 0.836 | 0.55343 |
| **Final A** | ≈ 0.440 | ≈ 0.817 | ≈ 0.994 | **0.88376** |
| **Final B** | ≈ 0.440 | ≈ 0.815 | ≈ 0.994 | **0.88324** |

Task values for A and B are derived from the leaderboard decomposition, not measured directly.

The final pipelines are:

- **Task 1 + 3.** A two-stage LightGBM residual model on top of temporal interpolation. It uses spatial, characteristic-line, ramp-flow and conservation features, a density model with reconciliation, specialist models for the 90-minute Task 2 horizon gap and ramp gaps, and a small self-supervised correction fitted inside each scored split on extra-masked public cells.
- **Task 2.** Per-cell LightGBM classifiers.
  - Onset windows: the classifiers predict the queue set at T+30. They mix generic features with bottleneck flow-threshold features.
  - Ongoing windows: a mix of four model families, followed by a lower decision threshold for windows with small queues at T.
  - Both finals mix these ongoing probabilities with a three-seed spatio-temporal CNN, at weight 0.5 (A) or 0.7 (B). A climatology gate keeps the CNN from adding a queue on a link that is empty at T where the train climatology says congestion is likely.
- **Task 4.** Regularised non-negative least squares on each split's own released counts (the organiser baseline used train counts). This is within about 0.006 of the S_ODME ceiling.

Main findings:

1. **S_physics is a density-accuracy problem, not a physics-model problem.** With the organiser's boundary flows, S_LWR reduces to an L1 error of the vehicle accumulation in masked cells. Conservation-based reconstruction does not help, because flow counting noise per five minutes is about as large as a link's accumulation (Section 3).
2. **Task 2 local scores do not transfer to the scored months.** Our out-of-fold S_queue was about 0.13 above the leaderboard, and many changes that helped locally lost on the leaderboard. We traced part of this to how much the row at T tells about the next 30 minutes in the scored months, and part to models leaning on train-month climatology. Changes that reduced climatology dependence were the ones that transferred (Section 4).
3. **The Task 4 baseline was limited by its input, not its solver.** Solving with the split's own counts raised S_ODME from 0.836 to about 0.994. Pulling the prior towards the cross-split structure lowered it to about 0.888 (Section 5).

Everything is rebuilt from the public release by one script (`final/run_all.sh`). The package lists the data, software and licences used. On macOS and Linux (both ARM), Task 2 is byte-identical and Task 4 differs only in the last floating-point digit. On one x86 machine an earlier Task 2 candidate differed in 53 of 174,000 cells. Task 1 agrees with the submitted file at score level on every machine we tried. The closest rebuild differs in 1.9 % of cells by a mean of 0.002 km/h (Section 6).

## 2. Scoring and validation

### 2.1 How the score is built

S_total = 0.35·S_state + 0.30·S_queue + 0.15·S_physics + 0.20·S_ODME.

Each task score is computed per panel (corridor direction), averaged over the two directions, and then averaged over the five corridors. The public and private leaderboards score different months (validation and private), and both months were generated with scenario seeds different from train. Submitting the ground truth scores 0.9945 because of the noise floor in Task 1 and Task 3.

Two properties shaped our whole approach:

- **The four task scores are additive and independent.** Task 3 depends only on the Task 1 rows, so Task 1 and Task 3 move together. Nothing else interacts.
- **The upper bound for each task is far from 1 in a known way.** Task 4 is near its ceiling after one change (Section 5). Task 1 + 3 is limited by measurement noise (Section 3). So most of the remaining room was in Task 2.

### 2.2 Leaderboard decomposition

We changed one task file at a time against a fixed reference and divided the score change by the task weight (0.30 for the queue, 0.20 for ODME, and 0.35·S_state + 0.15·S_physics together for the state file). Appendix A lists all 56 submissions with the reference each is compared against.

The decomposition is exact up to leaderboard rounding (five decimals), because the composite is linear in the task scores. It gave us four calibrated task scores from a single number.

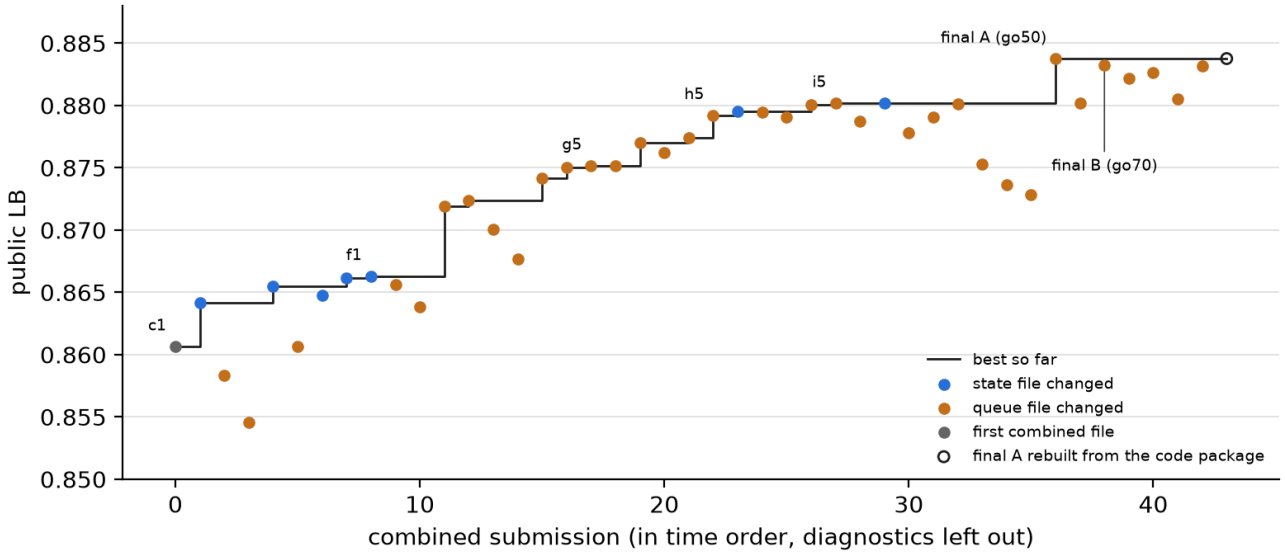



*Figure 1. Public LB of every combined submission, in time order (diagnostic files left out). Colour shows which task file differs from the reference. The line is the best score so far. The open circle is final A rebuilt from the code package (Section 6.2). Task 1 changes (blue) gave steady small gains. Most of the later movement, up and down, came from the queue file.*

Some submissions were diagnostics, not candidates:

| Submission | What it isolates | Result |
|---|---|---|
| q2_diag_onset | lgb_v1 on onset rows, persistence on ongoing rows | lgb_v1 onset ≈ 0.75, ongoing ≈ 0.77 on the LB (0.89 / 0.88 locally) |
| c9_diag_persT | ongoing = persistence of the row at T | ongoing persistence ≈ 0.669 on the LB (0.700 locally) |
| d9_diag_zeros | queue file all zeros | S_queue ≈ 0.000: empty predictions earn nothing in any window |
| d1, d3, d4, d5 | lgb_v1 on one group of corridors, T−5 persistence elsewhere | which corridors carry the local–LB gap |
| d10_diag_onsetdil | onset sets dilated by one link on each side | whether onset errors are near misses |

### 2.3 Local validation per task

| Task | Local proxy | Agreement with the leaderboard |
|---|---|---|
| 1 | The official Task 1 formula on train's real masks, held-out days (day index mod 9 = 4) | Good. gbm_v4c gained 0.1383 on the LB against 0.1428 predicted (its local C minus the baseline's leaderboard C of 0.2957); gbm_v8s2 gained +0.0035 against +0.0037 predicted. |
| 3 | S_FD exact; S_LWR proxy $1 - \Sigma\lvert\Delta N_{pred} - \Delta N_{obs}\rvert / \Sigma\lvert\Delta N_{obs}\rvert$ (Section 3) | Equal to the official evaluator to six decimals when it is fed observation-consistent boundary flows |
| 2 | Pseudo-windows on train that imitate the window selector (72 of 80 official train windows reproduced), then month-level out-of-fold scoring | About 0.13 too high in level. It often ranked candidates in the wrong order (Section 4) |
| 4 | S_link only; the other parts need the withheld path flows | Not informative beyond S_link |

For Task 1 we trusted the local proxy and used the leaderboard only to confirm. For Task 2 we used the local score only to filter. In the last week a change went to the leaderboard only if it passed a safety check on validation windows and had a reason to transfer that did not depend on train-month climatology.

## 3. Task 1 and Task 3: state reconstruction and physics consistency

Task 1 asks for speed and flow at every masked cell of the validation and private months. Task 3 has no submission of its own; the evaluator scores the Task 1 answer for physical consistency. We therefore optimise one combined contribution

```
C = 0.35·S_state + 0.15·S_physics = 0.35·S_state + 0.05·S_FD + 0.10·S_LWR.
```

The organiser terms:

- **S_state**: per masking regime r ∈ {R1, R2, R3} (mask rates 0.20, 0.30, 0.50), 0.54·max(0, 1 − RMSE_speed/25) + 0.46·max(0, 1 − RMSE_flow_per_lane/600), averaged over regimes, on score-eligible cells.
- **S_physics** = (1/3)·S_FD + (2/3)·S_LWR. The evaluator derives density k = q/v and accumulation N = k·L from the submitted state.
- **S_FD**: a per-lane fundamental-diagram validity check. It lies between 0.9862 and 0.9874 for every model below.
- **S_LWR** = max(0, 1 − Σ|residual| / Σ|rhs|) over five-minute transitions of N(t+dt) − N(t) = dt·(q_in + r_on − q_out − r_off), with organiser-derived boundary flows. All ten corridors are in coverage mode A.

### 3.1 Data layout, validation and baseline

Each panel is held as dense (day, 5-minute slot, link) arrays, links ordered by milepost. In all panels, flow perturbations travel towards increasing milepost (5-minute lagged correlation at 6–12 km: +0.30 to +0.36 from the lower-milepost side, −0.14 to −0.16 from the other), although the topology file points the other way for the five southbound and westbound panels. Reordering those panels to the topology direction lowered C from 0.43661 to 0.43548.

Local scores are computed on held-out train days (day index mod 9 = 4, about 30 days per panel, all regimes). Training rows come from the other days. S_state and S_FD use the official formulas. For S_LWR we use the proxy 1 − Σ|ΔN_sub − ΔN_obs| / Σ|ΔN_obs| (see the Task 3 subsection). Fed with observation-consistent boundary flows, the official evaluator matches it to six decimals (D12_I405_S, all regimes; R3 0.320715 against 0.320715).

| Method | S_state | S_FD | S_LWR | S_physics | C |
|---|---|---|---|---|---|
| Organiser baseline (weekday × time-of-day mean), D12_I5_N full train | 0.6255 (official 0.6252) | 0.9729 | 0.0000 | 0.3243 (official 0.3242) | 0.2676 (this panel) |
| Temporal linear interpolation of the same link, 10 panels, held-out days | 0.9084 | 0.9862 | 0.5262 | 0.6795 | 0.4199 |

All models below predict the residual against temporal interpolation, with flow per lane.

### 3.2 Model

Model names follow one line of development. gbm4 is a single-stage model on the v4 features (dataset ds4). gbm7 keeps those features but trains on 200,000 rows per regime with gap cells sampled four times (dataset ds7). gbm_v4c is gbm4 with clipping and reconciliation, and gbm_v8s2 is the two-stage model on ds7. The final v7 runs two stages on ds13 (ds7 rows plus ramp and conservation features), with ramp gap specialists and the three-mask self-supervised step (ss3f).

**Features.** The v4 set has 176 columns: own-link observations before and after the cell with their distances, ±2 and ±6 slot means and a local trend; neighbour links ±1 to ±3 at the same slot and at t ± 1, with their residuals; distance-weighted residual means at 0.5, 2 and 8 km, corridor-wide, upstream-only and downstream-only; characteristic-line residuals at x ∓ v·5 min for t ∓ 1 and t ∓ 2; and Task 2 window descriptors from the public `window_index.csv`.

The final version (v7) adds two groups from released data.

- **Ramp flows** (`ramp_states`): on- and off-ramp sums at the link and ±1, ±2 links, short means, observed share, ratio to a weekday × time profile, and cumulative net inflow Σ(on − off). Missing ramp values are filled by the ±2 slot mean, then the profile, never by zero.
- **Mainline conservation**: from the nearest observed upstream link u and downstream link d, an upstream estimate Q_u + Σ(on − off) and a downstream estimate Q_d − Σ(on − off), with their mean and difference per lane, the distances to u and d, and the observed ramp share between them. The mean estimate alone has flow RMSE 32.7, against 38.3 for the interpolation baseline and 180 for the raw neighbour flow (D7_I10_W held-out rows).

**Training rows.** Rows are drawn from train's real masks, 200,000 per panel and regime. Task 2 horizon gaps are imitated on train by reproducing the organiser window rules and blanking [T+5, T+90]; cells inside them are sampled four times. Three LightGBM (Ke et al., 2017) regressors predict speed, flow per lane and density per lane. The density model is weighted by lanes × link length, the vehicle units that S_LWR sums.

**Stage 1 and stage 2.** Stage 1 is fitted separately on odd and even days (lr 0.1, 800 rounds), so each half fills the other half's empty cells out of fold. Stage 2 learns the stage-1 residual from neighbour features of this filled field (about 6.2 M rows, about 280 columns, lr 0.05, three seeds averaged). Because early stopping depended on the thread count (3,348 rounds on 8 threads, 3,983 on 16), the final chain uses a fixed 4,000 rounds.

**Clipping and density reconciliation.** Speed is clipped to [link train minimum, 120.75 km/h], the generator's cap. Flow per lane is capped at 1.1 × the link's train maximum. The implied density q/v is then moved towards the density model, carried by flow only. The

step is computed in logs, after flooring v at 1 km/h, q at 1 veh/h/lane and k̂ at 0.001 veh/km/lane; the flow cap is applied after it:

```
q ← q^(1−β) · (k̂·v)^β,  β = 0.8 if predicted v < 80 km/h, else 0.2.
```

On gbm7, this traded S_state 0.9400 → 0.9377 for S_LWR 0.6092 → 0.6313, and C rose from 0.4393 to 0.4407. Moving speed as well cost more S_state than it gained.

**Gap specialists.** The Task 2 horizon gap blanks the whole mainline for 90 minutes after each forecast origin. A first specialist trained on gap cells only (gbm9gap, 1.18 M rows) blended at 0.5 added +0.0003. Inside these gaps 79 % of ramp flows are still observed. Task 1 has no causality constraint, so they are valid inputs; Task 2 does not use them. Four ramp specialists replaced gbm9gap: gap8 (v4 + ramp) and gap8b (v4 + queue-polygon + ramp), each with two seeds. Their mean is blended at 0.75 on the gap cells listed in `window_index.csv`. On the v4 stage 2 this added +0.000498. It was the first gap experiment to improve congested and free-flow cells together: gap speed RMSE fell from 16.00 to 13.58 (congested) and from 5.32 to 5.03 (free flow). The gain came entirely through speed.

**Self-supervised in-split correction.** For each scored split, 15 % of the visible eligible cells outside the gaps are hidden again and predicted by the full model. A small LightGBM model R (31 leaves, 300 rounds, 21 columns including link and panel as categoricals) learns the residual of speed, flow and density against their published observations. R's prediction is added to the template cells before clipping and reconciliation. No hidden value is used. This is allowed for Task 1 because it is an offline reconstruction: the visible cells of the scored months are released inputs, and the masked targets are never read (Section 7.2). Task 2 does not use this step. The final form (ss3f) runs three disjoint 15 % masks, which triples R's training cells with the same missing-neighbour context. The single-mask correction is small: RMS 0.65 km/h in speed and 17 vph in flow.

**Final configuration (v7).** Cache; dataset ds13 (ds7 rows plus ramp and conservation features); odd/even stage 1 and fill; stage 2 with three seeds × 4,000 fixed rounds; four ramp gap specialists at 0.75; ss3f; clipping and reconciliation. Deterministic rebuilds from code-only copies on a 20-core Linux aarch64 machine (16 threads) took about 13 h 40 min at about 20 GB peak memory. They differ from the submitted file in 1.9 % of cells, by a mean of 0.0020 km/h and 0.19 vph.

### 3.3 Results: local proxy against the public leaderboard

Each submission below changed only the state file against fixed queue and ODME files, so the LB change equals the change in C.

| Sub. | State candidate | Change | Local C | Local Δ | Public LB | LB Δ |
|---|---|---|---|---|---|---|
| c1 | gbm_v4c | stage 1 + clipping + reconciliation | 0.4385 | – | 0.86066 | – |
| c8 | gbm_v8s2 | two-stage model | 0.4422 | +0.0037 | 0.86414 | +0.00348 |
| d8 | gbm_v8s2_ss | + self-supervised correction | not measurable | – | 0.86552 | +0.00138 vs c8 |
| e1 | gbm_v8s2x3g_fixed | 3 seeds + gbm9gap 0.5, fixed rounds | 0.44290 (early-stopped run) | +0.0007 | 0.86478 | +0.00064 vs c8 |
| e2 | gbm_v8s2x3g_ss_fixed | e1 + self-supervised | not measurable | – | 0.86616 | +0.00138 vs e1 |
| f1 | gbm_v8s2x3g_ss3f_fixed | three disjoint masks | not measurable | – | 0.86631 | +0.00015 vs e2 |
| i1 | v7 | ramp stage 2 + ramp gap specialists | paired test 0.44179 → 0.44330 | +0.00151 | 0.87951 | +0.00029 vs h5 (0.87922) |
| j1 | rf7 | ramp columns added to R | not measurable | – | 0.88019 | +0.00000 vs i5 (0.88019) |

i1 and j1 use later queue files and are compared with the submissions that share them.

- **Model changes transfer.** Against the organiser baseline, gbm_v4c raised C by 0.1383 on the LB. Its local C of 0.4385 minus the baseline's leaderboard C (0.2957) predicted 0.1428. The two-stage model gave +0.0035 against +0.0037, and seeds plus the gap model +0.00064 against +0.0007.
- **The self-supervised step is invisible on train.** In a three-panel train test, R fitted on the evaluation month gave +0.0004 and R fitted on a control month +0.0007, so the gain there is generic second-pass correction. On the LB the same step gave +0.00138 twice, on two base models. We read this as evidence that the scored months differ in distribution from train.
- **v7 is the exception.** The proxy overstated it about five times. The local figure comes from a reduced paired test (150,000 rows per regime, stage 2 lr 0.1 × 1,000 rounds, one seed, no self-supervised step). Both sides of that test had a gap specialist (gbm9gap at 0.5 for the reference, the ramp specialists at 0.75 for v7), as on the leaderboard, so a missing specialist does not explain the gap. The leaderboard reference had the self-supervised step, which may already correct part of what v7 adds; we did not test this. We have no confirmed explanation. On the test template, 79.3 % of the 401 cells where v7 and ss3f differ by more than 2,000 vph are gap cells, against 1.9 % of all cells; on train the share is 26 %. This appears to be because test windows are real congestion windows, with more congestion inside the gap. On train, the v7-type model was the more accurate one on every such subset we checked.

### 3.4 Error analysis

The breakdown uses gbm4 with clipping and reconciliation (β 0.2/0.8) on about 3.5 M masked cells of the held-out days. Loss is the shortfall of C from a perfect answer, 0.0607, excluding the S_FD share of about 0.001. Speed and flow loss is assigned by squared-error share within panel and regime. Each S_LWR transition residual is split between the masked cells at its ends.

| Loss component | Share |
|---|---|
| S_LWR | 62.5 % (0.0380) |
| speed | 22.3 % (0.0136) |
| flow | 15.1 % (0.0092) |

| Traffic state (true speed, km/h) | Cells | Speed RMSE (km/h) | Flow RMSE (veh/h/lane) | Share of loss | of which S_LWR |
|---|---|---|---|---|---|
| free flow (≥ 95) | 95.4 % | 1.66 | 33.0 | 76.2 % | 49.2 % |
| congested (< 60) | 4.3 % | 5.11 | 77.2 | 21.6 % | 12.5 % |
| transition (60–95) | 0.3 % | 7.59 | 103.0 | 2.3 % | 0.9 % |

| Gap length (consecutive unobserved slots) | Cells | Speed RMSE | Flow RMSE | Share of loss |
|---|---|---|---|---|
| 1 | 32.3 % | 1.44 | 32.1 | 33.2 % |
| 2–3 | 41.6 % | 1.64 | 34.0 | 35.8 % |
| 4–11 | 23.0 % | 2.07 | 38.8 | 20.3 % |
| ≥ 12 | 1.1 % | 2.43 | 45.8 | 1.1 % |
| Task 2 horizon gap | 2.0 % | 7.50 | 86.1 | 9.6 % |

The loss sits in three places. Short free-flow gaps (1–3 slots) hold about three quarters of it, at a speed RMSE (1.3–1.4 km/h) equal to measurement noise ($\sigma \approx 1.45$ km/h); little is recoverable there. Task 2 horizon gaps are the most expensive per cell: 2 % of cells, 9.6 % of the loss. Congested cells inside them (speed RMSE 19 km/h, flow RMSE 213 veh/h/lane) account for half of that. By regime, R3 days carry 43.6 % of the loss on 49.4 % of cells; peak hours (13–20 h) carry 52.4 % on 33.3 %. Panels are even at 8.3–11.5 % each.

The later changes worked on these places. In the paired test, ramp and conservation features lowered free-flow flow RMSE outside the gaps from 31.3 to 30.5 and congested gap speed RMSE from 16.00 to 14.83, and lifted R3 S_LWR from 0.467 to 0.477. Across the earlier sequence, R3 S_LWR went 0.398 (gbm4) → 0.444 (+ reconciliation) → 0.461 (gbm7 + reconciliation) → 0.473 (two-stage + reconciliation).

### 3.5 Task 3: what S_LWR measures

The evaluator rebuilds the full state (submitted values at targets, observations elsewhere), computes $N = (q/v)\cdot L$ and checks conservation against organiser boundary flows. Those flows agree with the released observations, as the six-decimal match shows, so the right-hand side is close to the observed $\Delta N$ and

```
S_LWR ≈ 1 − Σ|ΔN_sub − ΔN_obs| / Σ|ΔN_obs|.
```

This is an L1 error of masked-cell accumulation, normalised by the small natural $\Delta N$. It rewards accurate density at masked cells, in vehicle units, which is why the density model is weighted by lanes × length and reconciliation moves q/v towards it. Over the model sequence the S_LWR proxy rose from 0.5262 to 0.6374 while S_FD stayed flat. On the scored months even the true state would not reach 1; we estimate about 0.96, because of noise in the boundary flows.

**Gate experiment (negative).** Before building a conservation-based or Newell cumulative-curve reconstruction (Newell, 1993), we compared, on train's unmasked layer, the actual $\Delta N = N(t+1) - N(t)$ with $dt\cdot(q_{l-1} - q_l)$ built from neighbouring observed flows (40 days):

| Panel | Σ\|ΔN\| / ΣN | Σ\|rhs\| / Σ\|ΔN\| (slot t) | same, mean of t and t+1 | correlation with ΔN (slot t / mean) |
|---|---|---|---|---|
| D12_I5_N | 0.038 | 19.9 | 14.4 | 0.10 / 0.01 |
| D7_I405_N | 0.051 | 22.6 | 16.2 | 0.13 / 0.03 |

The right-hand side is 15 to 20 times larger than the real $\Delta N$ and nearly uncorrelated with it. Neighbouring link flows differ by 107–163 veh/h on average, 9–14 vehicles per 5 minutes, while a link holds about 12 vehicles. Flow counting noise of about 3 % is as large as the accumulation the equation tracks. A Newell curve therefore picks up an error the size of the link's vehicle count after one step, and projecting masked density onto observed-flow conservation fits noise. We dropped both. Conservation still helps as a level estimate of flow (RMSE 32.7 against 38.3 above), because it uses flows, which are large relative to their 3 % noise, rather than their small difference.

### 3.6 What did not work

Changes in local C unless marked LB. The adoption threshold was +0.0005 (+0.0003 for late post-processing tests).

| Idea | Result | Note |
|---|---|---|
| Order links by topology direction | −0.00113 | seed rerun identical |
| Density loss L1 / median / Huber / unweighted L2 | C 0.4397 / 0.4397 / 0.4405 / 0.4406 vs 0.4407 | weighted L2 kept |
| Per-regime reconciliation β grid (75 settings) | +0.00015 | in-sample; flat optimum |

| Idea | Result | Note |
|---|---|---|
| Arithmetic instead of geometric q blend | −0.00002 | |
| Larger stage-2 trees (511 leaves) | +0.0002 | not adopted |
| Congestion weighting: speed v < 60 × 2.5 / flow 1/max(v,15) / both | −0.00004 / −0.00011 / −0.00015 | congested speed RMSE 5.69 → 5.08, free flow 1.58 → 1.68 |
| Characteristic-line baseline in gaps | 0.2 blend −0.00156 | gap speed RMSE 20.64 vs 6.57 |
| Shock-polygon and moving-frame gap features | at most +0.00014 | congested 17.2 → 15.0, free flow 5.32 → 5.8–6.0 |
| Bottleneck-threshold gap features | at most +0.00002 | overlaps ramp features |
| Snap congested speed to link median | at most +0.00005 | |
| Six instead of four ramp gap specialists | +0.000002 | |
| Temporal smoothing inside masked runs | +0.00005 | |
| Ramp and conservation columns in R (rf7) | LB +0.00000 | R's inputs already use them |

Congestion weighting helps congested cells but costs more on free-flow cells, which are 95 % of the cells. The gap-shape features fail because most gap loss comes from congestion that forms and clears inside the 90-minute gap and leaves no trace at T or T+95. The estimated queue shift was mostly zero: queues sit at bottlenecks and grow and shrink rather than travel. Ramp flows observed inside the gap were the first input that carried information about these events.

## 4. Task 2: queue forecasting

Each Task 2 window gives one panel (a corridor direction) up to its forecast origin T: the history rows T−60 to T−5 and the masked mainline row at T. The answer is the binary queue state (speed ≤ v_cut) of every link at the six horizons T+5 to T+30. Task 2 has eight panels on four corridors. The public leaderboard scores 80 validation windows (five per panel and condition) and the private leaderboard 80 April windows.

The two final configurations differ only in the queue file. Both use Task 1 v7 and Task 4 nnls_own.

| Final | Queue file | Public LB | S_queue (derived) |
|---|---|---|---|
| A | cpu3g_go50: i5 with ongoing probabilities mixed 0.5 with a 3-seed CPU-deterministic spatio-temporal CNN, under a climatology gate | 0.88376 | ≈ 0.817 |
| B | cpu3g_go70: the same blend with network weight 0.7 | 0.88324 | ≈ 0.815 |

Both build on i5 (lgb_onthr_sq31_pao1535, LightGBM only, 0.88019, S_queue ≈ 0.805). Sections 4.2 to 4.4 describe i5 and the network blend.

S_queue is derived by chaining queue-only score differences divided by the weight 0.30, starting from lgb_v1 at 0.760 (Section 2.2). For comparison, the organiser persistence baseline scores 0.302.

### 4.1 Task structure we exploited

**Two conditions.** In `queue_onset` windows the history has no eligible queue and a queue appears inside the horizon. The question is which links queue. In `queue_ongoing` windows a queue is already visible and we forecast how it grows, moves and clears over six horizons.

**The metric.** A window scores the IoU between the predicted and true sets of queued cells (link × horizon) over score-eligible cells. Scores are averaged window → (panel, condition) → condition → direction → corridor. Three consequences shaped the decisions:

- An empty prediction scores 0 whenever the truth has a queue. The all-zeros probe scored S_queue 0.000, so every scored window has a queue.
- IoU is a ratio of set sizes, so we decide sets, not cells. For onset we rank links by probability at T+30 and keep the top k that maximise the ratio-of-expectations approximation (in the spirit of Ye et al., 2012) $\Sigma_{i\le k} p_i / (k + \Sigma_{i>k} p_i)$. For ongoing, a fixed probability threshold beat top-k by 0.005 locally, because queued cells are strongly correlated.
- One window is a fifth of one (panel, condition) cell of the public score. The public S_queue therefore has a standard error of about 0.03.

**The window selector.** We rebuilt the selector from its public description and checked it against the 80 official train windows. It walks forward in time and takes the earliest origin that qualifies. Windows of either condition must be at least 360 minutes apart.

- Onset: no eligible queue in T−60 to T, and a queue in the horizon.
- Ongoing: an eligible queue at two or more of the 12 history steps, a queue in the horizon, and an IoU of at most 0.9 for the persistence forecast that copies the state at T.

The rebuilt selector reproduces 72 of the 80 official train windows (39 of 40 onset windows).

**Onset happens at T+30.** The selector takes the earliest qualifying origin, so the first queued step of an onset window always falls on the last horizon. The organisers confirmed that onset queues appear only at T+30. We set T+5 to T+25 to zero in onset windows and model only the T+30 link set.

The same mechanism shapes ongoing windows. 74% of out-of-fold ongoing windows are the first origin that qualified after the spacing gap. The change from the state at T is therefore concentrated in the last horizons. The IoU between the clean state at T and the state at T+h falls from 0.948 at h1 to 0.533 at h6. In D7_I210_E, 98% of ongoing windows are such first origins, and 38 of its 45 out-of-fold ongoing windows have no queue at T+30.

**Local validation.** We built pseudo-windows from the nine train months with the rebuilt selector. Labels are observed speed ≤ v_cut (0.6 × free-flow speed) on eligible cells. The row at T is masked at the release rates (20, 30 and 50%). Folds are three-month blocks (June–August, September–November, December–February). Climatology tables for a window come from the months outside its fold. The main yardstick is an official-style monthly out-of-fold set: five windows per panel and condition per month, 719 windows in total. T−5 persistence scores 0.3030 on the pseudo-windows against 0.3017 officially.

### 4.2 Rule compliance

A window's prediction uses only data stamped at or before T:

- its own history and the row at T;
- online features from complete earlier days of the same split and from the same day up to T−60;
- train-only tables: climatology, bottleneck thresholds and the fitted models.

`ramp_states` is not read. Each window is predicted on its own. Mixing weights and thresholds come from train out-of-fold scores and public leaderboard results. None is fitted on release data.

The deletion test (Section 7.1) reruns every one of the 160 validation and private windows from raw files with all rows stamped after T removed, 337,781,713 rows in total. For the LightGBM chain (h5, i5; Section 4.3), every network blend and gate variant (Section 4.4), the split-climatology candidates and the per-panel adaptive gate (Section 4.7), 0 of 174,000 cells changed.

### 4.3 LightGBM model

All families are per-cell binary classifiers: one onset model (link at T+30) and one ongoing model (link × horizon). They differ in the time-of-day information they see.

| Family | Features in plain terms | Used in i5 for |
|---|---|---|
| v1 | Speed / v_cut at T and over the history; flow / capacity; neighbour links; queue-tail position and extrapolated tail speed; time-of-day climatology; onset-location prior. One seed, 300 rounds. | onset 0.5, ongoing 0.15 |
| v1_thr | v1 onset features plus 8 bottleneck-threshold features (below) | onset 0.5 |
| phys | v1 state features without any time-of-day input, plus shock-wave tail extrapolation, backward-wave speed, capacity margin (own link, minimum over 6 downstream links), and a static onset propensity per link | ongoing 0.15 |
| onphys | phys plus online features from earlier days of the same split | ongoing 0.35 |
| aug | onphys plus time-of-day climatology. During training only, the climatology lookup time is drawn at random within ±90 min, so the model treats climatology as a rough prior. | ongoing 0.35 |

Feature details:

- **Shock wave.** The tail speed is $w = \Delta q/\Delta k$ between the 3 links above the tail and the 3 tail links (Rankine–Hugoniot), clipped to ±40 km/h, and extrapolated to each horizon.
- **Online features.** For up to 10 earlier days of the same split: queue frequency per link at the same clock time (T, T+30, T+60, ±15 min), and how far each link's first queue time lies from T+30. For the same day: whether a queue occurred before T−60, and how long ago.
- **Bottleneck thresholds.** Each bottleneck (Chen et al., 2004) has a hidden flow threshold. We estimate it from train days as the median per-lane flow just before the first daily queue. Folds use thresholds fitted on the other folds' days. Release windows use all train days. The features include the ratio of current flow to the threshold, its trend, the 30-minute maximum ratio, the remaining gap and the bottleneck's rank. They use only T−60 to T.

Phys, onphys, aug and v1_thr each average three seeds.

**Mixing and set selection.** Probabilities are mixed arithmetically before the decision. Logit averaging and geometric means looked better without a time shift (+0.0036 to +0.0041). Under shifted time-of-day inputs they lost 0.005 to 0.014, because they follow v1 whenever v1 is low. Onset keeps expected-IoU top-k. Ongoing uses threshold 0.37.

A candidate went to the leaderboard only if it passed three checks:

- out-of-fold no-shift $\Delta S_{queue} \geq -0.003$;
- a positive mean gain when all time-of-day inputs are shifted by ±45 and ±60 minutes;
- for onset changes, a guard on three low-confidence validation onset windows.

The guard came from the g2 failure (an onset mix of v1 0.6 and aug 0.4 that gained under time shifts locally but lost 0.0158 S_queue on the leaderboard; Section 4.6). Onset changes that removed links 59 and 61–63 from window D12_I5_N_val_004 were not submitted. The guard is itself informed by one leaderboard result, so we report it as a mild form of leaderboard selection. Each submission changed one setting.

**Small-queue threshold.** Windows with 1 to 5 queued links at T (empty cells at T filled from T−5) use threshold 0.31 instead of 0.37. There are 136 such windows among the 359 out-of-fold ongoing windows. On v1, ongoing windows with 15 or fewer queued cells lost 0.2 to 0.38 IoU, against an average of 0.103. The lower threshold gave +0.0015 ± 0.0009 out of fold. On release data it switched 56 cells, all from 0 to 1, and gained +0.00183 on the leaderboard. Stepped thresholds by queue size all lost locally (−0.0016 to −0.0055).

**The chain to i5.**

| Step | What changed | Public LB | ΔS_queue (derived) |
|---|---|---|---|
| lgb_v1 (f1) | start | 0.86631 | |
| ongphys | ongoing v1 0.5 / phys 0.5, threshold 0.37 | 0.87189 | +0.0186 |
| ongphys03 | ongoing v1 0.3 / phys 0.7 | 0.87240 | +0.0017 |
| g3 | ongoing v1 0.2 / phys 0.6 / aug 0.2 | 0.87419 | +0.0060 |
| g4 | v1 0.2 / phys 0.4 / aug 0.4 | 0.87505 | +0.0029 |
| ongpa26 = g5 | v1 0.2 / phys 0.2 / aug 0.6 | 0.87518 | +0.0004 |
| h2 | g5 + small-queue threshold 0.31 | 0.87701 | +0.0061 vs g5 |
| h4 | g5 + onset v1 0.5 + v1_thr 0.5 | 0.87738 | +0.0073 vs g5 |
| h5 | h2 + h4 | 0.87922 | +0.0135 vs g5 |
| i1 | state file v7 (not a queue change) | 0.87951 | |
| i5 | ongoing v1 0.15 / phys 0.15 / aug 0.35 / onphys 0.35 | 0.88019 | +0.0023 vs i1 |

The h2 and h4 effects added exactly: +0.00183 + 0.00220 = +0.00403, against +0.00404 measured. h4 changed only 12 cells in 7 onset windows, 2 of them in validation, so its leaderboard gain carries much luck. It was still the first onset change that improved the leaderboard.

**Leaderboard selection.** Several steps in this chain (ongphys03, g4, g5, i5) are below 0.003 S_queue, which one window can move on its own (Section 4.5). We do not claim them as separate gains. Each passed the local checks above before it was submitted, and together they move one way: the v1 share of the ongoing mix never rises (1.0 → 0.5 → 0.3 → 0.2 → 0.15), moving weight towards families that use no time-of-day input (phys, onphys) or use it only as a jittered prior (aug). The queue-only chain from lgb_v1 to i5 gained about +0.045 S_queue in total (+0.0430 from f1 to h5, +0.0023 from i1 to i5), most of it in the large steps (ongphys, h2, h4). The private month will show how much of the small steps was selection noise.

### 4.4 The network

**Inputs.** The network sees only the window's 13 rows (T−60 to T) and train-only tables.

- Dynamic channels [7 × 13 × L]: speed / v_cut (filled along time), observed flag, flow / capacity, occupancy, observation rate, eligible observed queue, filled queue.
- Per-link channels [26 × L]: 5 static link attributes; climatology queue probabilities at T, T+15, T+30 and T−30, the corridor maximum, the onset-location prior and the onset frequency; time of day (sin, cos); weekend; condition; 10 bottleneck-threshold features computed as in the LightGBM model.
- A learned 8-dimensional embedding per link.

**Architecture.** The network has about 810,000 parameters.

1. Two 2D convolutions (3 × 5) over time × link.
2. A convolution spanning all 13 rows, which compresses the time axis, joined with the features of the row at T.
3. Eight residual 1D convolution blocks along the link axis, with dilations 1, 2, 4, 8 applied twice. The receptive field is about ±120 links.
4. Six horizon logits per link. For onset windows only T+30 is predicted.

**Loss and training.** The loss is masked BCE on eligible cells plus a per-window soft IoU with weight 1.0. Training uses every train origin that meets a condition: 2,328 onset and about 68,000 ongoing origins, with onset making up 30% of each batch.

- For each sample the mask on the row at T is drawn again at the release rates.
- With probability 0.7 the time-of-day lookups are shifted by up to ±60 minutes.
- Training runs 12,000 steps with batch 96 and AdamW (learning rate 2e-3, one-cycle, weight decay 1e-4).
- There are three seeds. The third uses uniform panel weights and adds 30% of windows above the persistence-IoU cap.

**CPU-deterministic mode.** The final weights were trained on the CPU of a 20-core Linux aarch64 machine. We used `torch.use_deterministic_algorithms(True)`, fixed seeds, numpy generators for sampling and 2 threads. Each seed took about 4 h 45 min, with the three seeds running at the same time.

- Two runs with the same settings gave identical weight files. A different thread count changes the hash (mean probability difference 2e-8), so the thread count is fixed.
- Inference on another machine changed probabilities by at most 1.8e-5 and gave the same submission file. Training on another machine did not reproduce bit for bit (losses at 500 steps 0.5970 against 0.6005), so we ship the weights.
- GPU runs were used for exploration only. On the submission metric the CPU and GPU go50 blends differ by 1.21%, less than the 1.72% from changing one seed.

**Out-of-fold result.** The network alone scores like LightGBM (S_queue 0.882 to 0.910 per seed). Its cell probabilities correlate with LightGBM's at 0.980 out of fold and 0.973 on release data. Mixed 0.5 into the i5 ongoing probabilities, it gained +0.0132 ± 0.0025 over i5. The mix improved 213 windows and worsened 81, and it improved all nine months (+0.005 to +0.020). Mixing into onset at weight 0.2 gave +0.0027 locally but failed the onset guard, so onset stays LightGBM only. Five seeds did not improve on three (go50 −0.0003).

**Why the gate was needed.** The ungated blend lost on the leaderboard at every weight: go30 0.87905, go50 0.87876 and go70 0.87778, that is −0.0038, −0.0048 and −0.0080 S_queue against i5. We resampled the per-window out-of-fold differences 10,000 times in the public layout (five onset and five ongoing windows per panel). The chance of a result as low as the leaderboard was 0.24%, 0.15% and 0.18%, and 0.05% for all three at once. We found no leakage: checkpoints are not selected, and the fold and full models agree to 0.003 mean probability. The cause was in the kind of cell the network changed:

| Cell type (state at T, train climatology at T+h) | Direction | Share of changes out of fold | Correct out of fold | Share in validation | Share in private |
|---|---|---|---|---|---|
| Empty link, climatology ≥ 0.5 | 0 → 1 | 6.1% | 50% | 42.6% (172 cells) | 12.8% |
| Empty link, climatology < 0.1 | 0 → 1 | 14.7% | 76% | 16.6% | 7.1% |
| Queued link, climatology 0.1–0.5 | 1 → 0 | 18.3% | 81% | 12.6% | 23.3% |
| Other | | 60.9% | 60–90% | 28.2% | 56.8% |

The network trusts climatology more than the i5 mix, whose ongoing part draws little on time-of-day features. It adds queues on links that are empty at T but usually queued at that clock time. Even out of fold these additions are right only half the time. In validation they make up 43% of the network's changes. Climatology is also less reliable there (Section 4.5).

The gate applies to ongoing cells where the link is free at T (row at T, empty cells from T−5) and the train climatology at T+h is at least c. In those cells the mix may not turn on a queue that i5 left off, so the cell keeps i5's decision. With c = 0.5 the gate blocked 209 cells. It cost almost nothing out of fold (+0.0131 instead of +0.0132) and cut the validation cells that differ from i5 from 404 to 239.

We fixed the acceptance rule before the submission: accept go50 as an alternative if its score was at least 0.88019. The gated blend scored 0.88376. That is +0.00357 over i5 and +0.0050 over the ungated blend. Further gate variants are in 4.6.

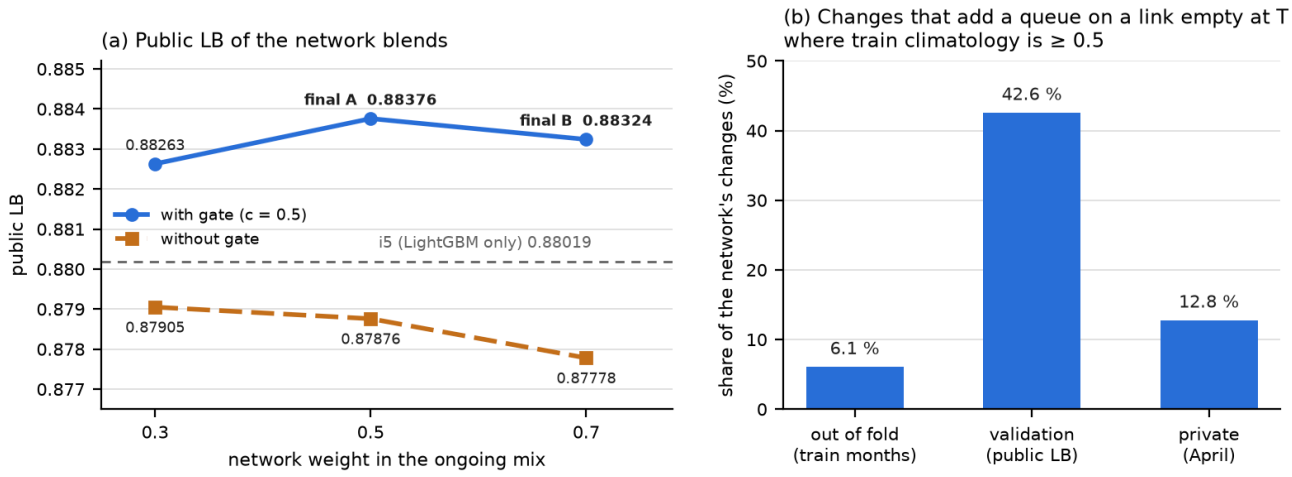



*Figure 2. (a) Without the gate the network blend scores below i5 at every weight; with the gate it scores above i5 at every weight, highest at 0.5. (b) The cells the gate blocks are 6 % of the network's changes out of fold but 43 % in validation, where train climatology is less reliable (Section 4.5). In the private month they are 13 %.*

**Choice of B.** The award pool is the top ten eligible teams, so the second final should score close to A in expectation while still differing from it on the private windows. We set the rule "public score within 0.002 of A" before comparing candidates. On the private file, i5 differs from A in 266 cells over 35 windows but scores 0.0036 lower, so it fails the rule. Among the gated variants that pass, go70g is the closest to A (0.0005 lower) and differs in 95 cells over 24 windows. go30g differs in about as many (107 cells, 27 windows) but scores 0.0011 lower. ga50 and c00 differ in only 23 and 30 cells, so they would add almost nothing as a second pick. B is therefore go70g. It shares A's gate, so the pair is not a hedge against the gate itself.

### 4.5 The local-versus-leaderboard gap

lgb_v1 scored 0.886 on the pseudo-windows and 0.760 on the leaderboard. Every model we built kept a gap of roughly this size.

**Ruled out:**

- Submission path and merge: the official train windows predicted through the submission code give the same output, and the uploaded values equal the candidate file in all 174,000 cells.
- Label noise: only 1–4% of horizon queue cells have a speed ratio of 0.9 to 1.0, and noise-aware threshold tuning left the decision rule unchanged.
- v_cut variants, demand level on free-flow links, bottleneck locations, the share of missing cells at T (0.42 against 0.44), condition definitions, and links without detectors (probe d1 below).
- Window-size mix: reweighting to the validation size distribution still predicts 0.872.
- A single time shift. Shifting time-of-day inputs by ±45 minutes reproduces the leaderboard pattern locally. But models without time of day did not gain on the leaderboard (lgb_notod −0.0077, lgb_robust −0.020), and a per-split shift estimate came out as −50 to +40 minutes per panel with little gain in fit.

**Diagnostic submissions.** We compare against c1 (0.86066, S_queue 0.760) unless noted.

| Probe | Queue file | Public LB | Derived reading |
|---|---|---|---|
| b0 | organiser T−5 persistence (vs all-baseline) | 0.55343 | S_queue 0.302; local 0.303 |
| q2 | lgb_v1 onset, persistence ongoing (vs all-baseline) | 0.66649 | lgb_v1 onset ≈ 0.754, ongoing ≈ 0.766; local 0.896 / 0.877 |
| c9 | lgb_v1 onset, ongoing = persistence of the row at T | 0.84604 | T-row persistence ≈ 0.669 (local 0.700); T−5 persistence ≈ 0.603 (local 0.587) |
| d1 | lgb_v1 on the 4 sensor-only panels, persistence elsewhere | 0.78677 | S_queue 0.514: 46% of the gain (local 48.6%) |
| d3 / d4 / d5 | lgb_v1 on I5 + I10 / I210 / I405, persistence elsewhere | 0.77700 / 0.79194 / 0.78545 | LB/local gain ratio: I5 0.50, I10 0.74, I210 1.03, I405 0.87 |
| d9 | all zeros | 0.63274 | S_queue 0.000 |
| d10 | lgb_v1 with onset sets dilated by one link | 0.81607 | onset 0.457; ratio to undilated 0.606 (local 0.604) |

What the probes show:

- **The gain over persistence shrinks to 55–60% of its local value.** For the T-row persistence it falls from +0.113 to +0.066. For lgb_v1 it falls from +0.291 to +0.163. A mixture fits all three numbers: in about 40% of scored cells every predictor scores about 0.62, and the rest behave as locally.
- **Onset sets are not misplaced.** Dilation hurts on the leaderboard exactly as it does locally. A second mixture fits both onset numbers: about 84% of onset windows behave as locally and about 16% score zero whatever is submitted. Locally only about 5% of onset windows score zero.
- **The loss sits in D12_I5.** It is not a detector effect.

**What explains part of it.** The organiser documentation says the validation and private months have their own demand draws and incident schedules. The nine train months share one demand draw, so out-of-fold validation cannot see a change of demand pattern.

- **Climatology is less reliable in the scored months.** P(queue | climatology ≥ 0.5) on history cells is 0.765 out of fold, 0.709 in validation and 0.678 in private. For D12_I5_N it is 0.918, 0.536 and 0.404.
- **Validation D12_I5_N is a low-demand month.** Its weekday queue share is 2.30%, below the train minimum of 3.01% (mean 4.37%).
- **The zero-score onset windows have no precursor in the inputs.** They are weekday afternoons on days whose usual morning bottleneck never activated, or early-morning incident-like queues.
- **The model already knows where it is weak.** lgb_v1's own expected IoU falls in validation in the same corridor order as the leaderboard loss (validation/train ratio: I5 0.82, I10 0.85, I210 0.99, I405 1.03). The private ratio is 0.98 overall.
- **Smaller factors.** Atypical, incident-like congestion is more common in the scored months but explains only about 0.016. The row at T is less useful there: T-row persistence loses 0.031 against local while T−5 persistence gains 0.016.

**Where local and leaderboard disagreed in direction.** The table shows paired local and leaderboard changes. Local is the out-of-fold ΔS_queue against the same reference unless noted.

| Change | Local | LB ΔS_queue | Same sign |
|---|---|---|---|
| ongoing v1 → v1 0.5 / phys 0.5 | +0.0024 (vs v1) | +0.0186 | yes, 8 times larger |
| ongphys → ongphys03 | lower (−0.0019 vs v1, against +0.0024) | +0.0017 | no |
| g5 aug → onphys (h3) | −0.0006 | +0.0035 | no |
| small-queue threshold 0.31 (h2) | +0.0015 | +0.0061 | yes |
| onset + v1_thr (h4) | +0.0014 | +0.0073 | yes |
| ongoing v1 .2 / phys .2 / aug .3 / onphys .3 (i2, pao33) | +0.0014 | −0.0003 | no |
| horizon-wise mix (i4, hmix) | −0.0009 | +0.0018 | no |
| ongoing aug + thresholds (i3, athr) | +0.0015 | −0.0015 | no |

| Change | Local | LB ΔS_queue | Same sign |
|---|---|---|---|
| base-rate shift (f2, br03) | +0.0015 | −0.0024 | no |
| onset v1 0.6 + aug 0.4 (g2) | −0.0004 no shift, +0.045 under shift | −0.0158 | no |
| ungated net go30 / go50 / go70 | +0.0093 / +0.0132 / +0.0154 | −0.0038 / −0.0048 / −0.0080 | no |
| split climatology sca75 / scr75 | March test +0.0113 / +0.0086 | −0.0218 / −0.0164 | no |
| gated go70 vs go50 | +0.0022 | −0.0017 | no |
| gate c 0.7 vs 0.5 (c07) | +0.0000 | −0.0107 | local blind |
| gate c 0.0 vs 0.5 (c00) | −0.0030 | −0.0054 | yes |
| gated go30 vs go50 | −0.0041 | −0.0038 | yes |

The pattern is consistent. Local validation rewards anything that leans on train climatology, because train months share the climatology. The leaderboard punishes it. c07 shows this most clearly: it lets back exactly the cells that out-of-fold validation cannot judge. It is neutral locally and −0.0107 on the leaderboard.

We also scored the lowest-demand 30% of train days with models trained on the other days. That test matched the leaderboard direction for 7 of 11 changes, including all 3 onset changes. We used it only as a secondary check for onset.

After 10/29 we therefore chose by the leaderboard, one change per submission, with acceptance rules fixed in advance for the larger moves (the gate, split climatology). Each public window carries 1/80 of S_queue, so a step below about 0.003 S_queue can come from one window changing by a quarter of its IoU. Such steps should not be read as real.

### 4.6 Results: every queue-only change that reached the leaderboard

ΔS_queue = Δ(public LB) / 0.30 against the reference shown. The state and ODME files are identical within each comparison. Leaderboard scores are rounded to five decimals, so |ΔS_queue| < 0.0002 is noise.

| Submission | Queue change | Reference | Public LB | ΔS_queue |
|---|---|---|---|---|
| q1_lgb_v1 | lgb_v1 | b0 0.55343 | 0.69083 | +0.4580 |
| c3 | lgb_notod, threshold 0.35 | c1 0.86066 | 0.85836 | −0.0077 |
| c4 | lgb_robust (online onset + notod ongoing) | c1 | 0.85454 | −0.0204 |
| d7 | lgb_v2 | c8 0.86414 | 0.86067 | −0.0116 |
| f2 | lgb_v1 + base-rate logit shift α 0.3 (br03) | f1 0.86631 | 0.86559 | −0.0024 |
| f3_todsnap | lgb_v1 + onset time-of-day snap (todsnap) | f1 | 0.86381 | −0.0083 |
| f3_ongphys | ongphys | f1 | 0.87189 | +0.0186 |
| f4 | ongphys03 | ongphys 0.87189 | 0.87240 | +0.0017 |
| g1 | ongoing phys only, threshold 0.33 | f4 0.87240 | 0.87003 | −0.0079 |
| g2 | onset v1 0.6 + aug 0.4 | f4 | 0.86765 | −0.0158 |
| g3 | ongoing v1 .2 / phys .6 / aug .2 | f4 | 0.87419 | +0.0060 |
| g4 | ongoing v1 .2 / phys .4 / aug .4 | g3 0.87419 | 0.87505 | +0.0029 |
| g5 | ongoing v1 .2 / phys .2 / aug .6 | g4 0.87505 | 0.87518 | +0.0004 |
| h1 | g5, aug → 9-model multi-shift aug | g5 0.87518 | 0.87514 | −0.0001 |
| h2 | g5 + small-queue threshold 0.31 | g5 | 0.87701 | +0.0061 |
| h3 | g5, aug → onphys | g5 | 0.87622 | +0.0035 |
| h4 | g5 + onset v1 .5 / v1_thr .5 | g5 | 0.87738 | +0.0073 |
| h5 | h2 + h4 | g5 | 0.87922 | +0.0135 |
| i2 | ongoing v1 .2 / phys .2 / aug .3 / onphys .3 (pao33) | i1 0.87951 | 0.87943 | −0.0003 |
| i3 | ongoing aug → aug + thresholds (athr) | i1 | 0.87906 | −0.0015 |
| i4 | horizon-wise ongoing mix (hmix) | i1 | 0.88006 | +0.0018 |
| i5 | ongoing v1 .15 / phys .15 / aug .35 / onphys .35 | i1 | 0.88019 | +0.0023 |
| j2 | i5 + horizon-wise deviations | i5 0.88019 | 0.88013 | −0.0002 |
| k go30 / go50 / go70 | ungated net blend 0.3 / 0.5 / 0.7 | i5 | 0.87905 / 0.87876 / 0.87778 | −0.0038 / −0.0048 / −0.0080 |
| n scr75 | split climatology, retrained, λ 0.75 | i5 | 0.87527 | −0.0164 |
| n sca75 / sca5 | split climatology, features only, λ 0.75 / 0.5 | i5 | 0.87366 / 0.87285 | −0.0218 / −0.0245 |
| m1 sclim75p | split climatology for private onset only; validation equals i5 | i5 | 0.88019 | 0.0000 |

| Submission | Queue change | Reference | Public LB | ΔS_queue |
|---|---|---|---|---|
| m go50g (A) | net 0.5 + climatology gate 0.5 | i5 | 0.88376 | +0.0119 |
| p go70g (B) | net 0.7 + gate 0.5 | go50g 0.88376 | 0.88324 | −0.0017 |
| p c00 | net 0.5, gate blocks every new queue on links empty at T | go50g | 0.88215 | −0.0054 |
| m go30g | net 0.3 + gate 0.5 | go50g | 0.88263 | −0.0038 |
| p c07 | net 0.5, weaker gate 0.7 | go50g | 0.88055 | −0.0107 |
| q ga50 | net 0.5, per-panel adaptive gate | go50g | 0.88315 | −0.0020 |

The diagnostic probes (q2, c9, d1, d3, d4, d5, d9, d10) are in 4.5. Weight 0.5 and gate 0.5 form a peak in both directions. Every step away from them lost.

### 4.7 What did not work

Out-of-fold numbers are ΔS_queue on the 719-window set against the stated baseline.

| Attempt | Result |
|---|---|
| **Split climatology**: all climatology inputs blended λ·train + (1−λ)·same-split earlier days, validation windows included | March test with the production rule: +0.0113 (λ 0.75), +0.0109 (0.5). Out of fold: +0.0004 / −0.0100. LB −0.0218 / −0.0245 S_queue. The retrained version scr75 (+0.0086 March, +0.0031 out of fold) scored LB −0.0164. All three failed the onset guard. The advance rule (any ≥ i5 + 0.001) failed, so we stopped this line. |
| **Onset mixes** with aug, onphys, time-jittered v1, three-way threshold mixes, online thresholds, no-climatology v1_thr, or the network | g2 (v1 .6 + aug .4): LB −0.0158 despite +0.045 under shift. v1 .5 + onphys .5: −0.0047 out of fold. All later onset mixes passed the local test but removed links 59 and 61–63 in D12_I5_N_val_004 and were not submitted. |
| **Selector refit**: recalibrate ongoing probabilities by whether T−5 would also have qualified, and by horizon | −0.0034 (73 windows better, 109 worse); horizon only −0.0018. The probabilities already carry this information. Replacing early horizons with persistence: h1 −0.0070 up to h1–h5 −0.051. |
| **Month-ahead recalibration**: pick decision settings on month m, apply them to month m+1 | +0.0014 ± 0.0012 (6 of 8 month pairs up), with the true labels of month m. Chosen settings jumped between months. Validation has no labels, so the real gain would be lower. |
| **Month-ahead refit**: add the previous month at weight 3 and retrain all six model families | +0.0020 (SD 0.0075, SE 0.0026), 5 of 8 pairs up. Onset only +0.0034; ongoing only −0.0015. |
| **br03**: per-split base-rate logit shift, α 0.3 | +0.0015 locally, LB −0.0024. |
| **todsnap**: move time-of-day lookups for onset windows with expected IoU < 0.35 | 36 cells changed. LB −0.0083. |
| Causal model to fill missing cells in the row at T | Fill IoU 0.939 → 0.968, but S_queue −0.0024 ± 0.0025 |
| Morphological closing / island removal | −0.019 / within ±0.0006 |
| Anchoring early horizons to the queue at T | −0.004 to −0.006 |
| Limiting upstream growth per 5 minutes | −0.026 to −0.028 |
| Higher thresholds for T+15 to T+30 (0.42 to 0.52) | −0.0003 to −0.0024 |
| Switching on a second probability peak or the next usual bottleneck in onset | −0.001 to −0.028 |
| Ramp inflow surge before T as an onset signal | AUC 0.491; top-k IoU −0.018 |
| Test-time averaging of time-of-day lookups (±30, ±45 min) | 0 to −0.0012 |
| LambdaRank for onset; multi-horizon speed regression for ongoing | 0.856 vs 0.8705 (binary); 0.783 vs 0.857 (classifier) |
| Three v1 seeds; five network seeds instead of three | −0.0012 to −0.0001; −0.0003 |
| Multi-shift aug (3 shift seeds × 3 tree seeds) | +0.0005 locally, LB −0.0001 |
| Per-panel adaptive gate | −0.0003 out of fold, LB −0.0020 vs go50g |

## 5. Task 4: OD and path-flow estimation

### 5.1 Structure of the problem

| Item | Value |
|---|---|
| Paths per panel | 630 (I405_S) to 5,050 (I5_S, I10_W): one per zone pair o < d |
| Measured links | mainline only, 35 to 350 per panel; ramps and connectors have no counts |
| Independent constraints | about the number of zones (matrix rank 54–157); links in the same section give identical rows |
| Count definition | mean mainline `flow_vph` over 15–18 h; averaging the released rows directly matches within ±1.3 % |
| Feasibility | in every split, `A f = c` can be met exactly with non-negative flows (residual 1e-16) |

A path (o, d) crosses every cut between o and d. A measured count is the cut total $\Sigma_{o \le k < d} f_{od}$, so the difference between neighbouring cut counts is only "departures minus arrivals" of one zone. With thousands of unknowns and a few dozen constraints, the prior decides almost all of the answer.

We fitted the log of the released weak prior with origin effects, destination effects and a polynomial in log distance. This explains about 74 % of its variance ($R^2 \approx 0.74$). The residuals have variance 0.34–0.41 and correlate across splits at about 0.05. So every split's prior is a shared structure times independent multiplicative noise ($\sigma \approx 0.6$). The prior also strongly under-weights the boundary zones: in I405_S, departures from Z0 sum to 1,327 against a first-link count of 6,939.

### 5.2 What we found

The organiser baseline solves a stacked, prior-regularised NNLS (a generalised least squares form, Cascetta, 1984) with $\lambda = 0.05$ using **train** counts for every split. That is why its S_link is only 0.965.

| Candidate | Change | Local S_link (val / priv) | Public LB (other tasks at baseline) | S_ODME (derived) |
|---|---|---|---|---|
| baseline | NNLS $\lambda = 0.05$, train counts | 0.9646 / 0.9636 | 0.55343 | 0.836 (official) |
| **nnls_own** | same NNLS, the split's own counts and prior | 0.9999 / 0.9999 | 0.58501 (+0.03158) | **≈ 0.994** |
| nnls_w75 | as nnls_own, prior seed pulled 25 % towards the cross-split structure | 0.9999 / 0.9999 | 0.56381 (+0.01038) | ≈ 0.888 |

- **The input was the limit, not the solver.** With the split's own counts the constraints are met exactly. The distribution over paths then follows that split's prior, and S_ODME reaches about 0.994. The remaining room is at most about +0.001 on the total, so we fixed Task 4 here and spent no more submissions on it.
- **Denoising the prior was wrong (negative result).** In a held-out-cut cross-validation (drop 25 % of measured cuts, predict them), the prior pulled 25 % towards the shared structure was the best seed: mean relative error 0.0365, against 0.0384 for the split's own prior. On the leaderboard the same change cost 0.106 of S_ODME. The reference path flows keep the split-specific prior noise. A cut-total metric therefore cannot stand in for the path-level score.
- **Additive vs multiplicative projection.** A KL (entropy) projection of the prior (Van Zuylen and Willumsen, 1980) onto the counts also met the counts exactly. In held-out-cut validation it was slightly worse than NNLS (0.0486 vs 0.0384). We did not submit it once nnls_own was confirmed near the ceiling.
- **Ramp counts.** The ramp flows in `ramp_states` do not balance the mainline change (I405_S, 15–18 h: on-ramps 3,980, off-ramps 2,197, mainline change −300). No ramp-to-zone table is released, so we did not use them as zone totals.

Runtime: 2.5–6 minutes with three processes. The result is deterministic. Across machines it is identical up to the last floating-point digit (max |Δ path_flow| 2.0e-11).

## 6. Reproducibility

### 6.1 Entry point and environment

The code package (`sortdev_tfb.zip`) rebuilds both final submissions with one command:

```
PYTHON312=/path/to/python3.12 ./setup_env.sh
TFB_REL=/path/to/kaggle_public final/run_all.sh out
```

- Run order: Task 4, then Task 2 (all LightGBM models trained from the release in an empty folder), then the two Task 2 network blends (A and B), then Task 1, then merge and format check.
- Environments:
  - Main: Python 3.12 with numpy 2.5.4, pandas 3.1.0, pyarrow 26.0.0, scipy 1.18.1, LightGBM 4.7.0 (CPU) and polars 2.0.0.
  - Network: Python 3.12 with torch 2.14.1 (CPU only).
- Inputs: no external data, no pretrained models, no paid services. The only binary input that ships is the Task 2 network weights (9 MB). `NN_TRAIN=1` retrains them.

- Determinism:
  - Every seed is fixed in code.
  - LightGBM runs with `deterministic=True` and `force_row_wise=True`, and uses fixed round counts. Early stopping made Task 1 depend on the thread count (3,348 against 3,983 rounds for one model on 8 and 16 threads), so we removed it.
  - The network was trained on CPU in deterministic mode.

### 6.2 Runs from code-only copies

| Machine | Task 4 | Task 2 (A and B) | Task 1 | Note |
|---|---|---|---|---|
| Linux aarch64, 20 cores, 16 threads (final package zip, empty folder) | 2.5 min; last-digit float differences from the Mac file (max 1.6e-11) | 44 min LightGBM + 5 min per network blend; A and B **byte-identical** to the submitted files | 13 h 41 min, peak 20.1 GB; byte-identical to an earlier rebuild on the same machine; 1.9 % of cells differ from the submitted file, mean \|Δ\| 0.0020 km/h and 0.19 veh/h | rebuilt A scored **0.88376** on the public leaderboard, equal to the submitted A. The submitted Task 1 combined models trained here and on a Mac; this rebuild trains all of them here |
| macOS arm64, M4 Pro, 10 threads | byte-identical | **byte-identical** | about 6.5 h; mean \|Δ\| 0.095 km/h and 7.3 veh/h (p99 0.65 km/h, 66 veh/h) | LightGBM takes a slightly different path from stage 1 on |
| macOS arm64, M4, 24 GB RAM, 8 threads (final package zip, empty folder) | byte-identical (4.5 min) | 39 min LightGBM + 5 min network blend; A and B **byte-identical** to the submitted files | 24 h 11 min with heavy swapping; mean \|Δ\| 0.092 km/h and 7.1 veh/h from the submitted file (p99 0.62 km/h, 65 veh/h), the same size as on the M4 Pro | merged A and B pass the format check. 24 GB is enough but slow; see the memory note below |
| Windows x86, i7-10870H | max \|Δ\| 2.0e-11 | an earlier candidate (lgb_aug): 53 of 174,000 cells differ (threshold-edge cells, ARM vs x86 floating point; estimated effect ≤ 0.001) | not run | |

How large is the Task 1 difference in score terms? The change from the previous state model (ss3f) to v7 moved the public LB by +0.00029, with mean \|Δ\| 0.174 km/h and 25.3 veh/h. The largest cross-machine difference above is about half of that. So the expected score gap is of order 0.0002–0.0003. The 1 % reproduction rule allows about 0.0088.

Peak memory: about 21 GB for Task 1, 8 GB for Task 2 and 1 GB for Task 4. We recommend 32 GB of RAM. On a 24 GB machine Task 1 still finishes, but swapping stretches it to about a day.

### 6.3 Use of AI tools

We used an AI coding assistant (Claude Code, Anthropic) throughout the competition. It wrote and refactored most of the code, ran and logged the experiments, and drafted this report. The submitted pipeline contains no language model and calls no external service; the assistant was a development tool only. Every number in this report comes from the run logs and the submission ledger in the package. The author set the direction, checked the results and is responsible for the submission.

## 7. Rule compliance

### 7.1 Task 2: no data after the forecast origin

The rule is that a Task 2 prediction for a window may use only data stamped at or before its origin T. This covers every file: mainline rows after the blanked buffer, and `ramp_states` values inside the horizon. Our Task 2 pipelines use:

- the window's own history and the released row at T;
- online features from complete earlier days of the same split and from the same day up to T−60 (`online_features`);
- train-only tables: time-of-day climatology, bottleneck flow thresholds and the fitted models.

Mixing weights and thresholds were chosen from train out-of-fold scores and public leaderboard results, never fitted on release data. Each window is predicted on its own, and no other window's origin is read.

We tested this directly. For each of the 160 validation and private windows, we deleted every released row stamped after T and predicted that window again from the truncated data. In total we deleted 337.8 M rows.

- Cached split-level arrays were not used.
- Every array was checked (assert) to contain no row after T.
- Any function that reads split-wide statistics was set to fail if called.

Results:

| Part | Test | Cells changed |
|---|---|---|
| LightGBM chain (h5, i5) | `task2/trunc_test.py` | 0 of 174,000 |
| Network blends as submitted: go50 = final A, go70 = final B | `task2_nn/trunc_final.sh` (`build_mix.py --trunc --compare`); the 160 network input arrays rebuilt from truncated data were asserted equal to those used for the submission | 0 of 174,000 each |
| Other network blends (ungated go30/50/70, adaptive gates) | `task2_nn/trunc_all.sh`, `build_mix.py --gate-adaptive --trunc` | 0 of 174,000 each |

### 7.2 Other rules

- Data: only the competition release. Software: open-source Python packages, pinned in the package.
- No pretrained models, external datasets or paid services. The organiser repository (MIT) ships unchanged inside the package.
- Task 1 uses each scored split's own public observations for the self-supervised correction (Section 3). These are released inputs, not targets, and the masked target cells are never used.
- Task 4 uses each split's own released counts and prior.

## 8. Lessons and limits

- **Decompose the leaderboard before tuning against it.** Changing one task per submission turned the composite into four calibrated task scores. It also showed early where the room was: Task 4 needed one change, Task 1 + 3 was close to its noise floor, and Task 2 held most of the remaining gap.
- **Check a physics assumption against the data before building on it.** Conservation looked like the natural tool for Task 3. A one-day gate experiment showed that counting noise is as large as the signal at five-minute resolution, which saved the effort of building a conservation solver.
- **A proxy that agrees with the leaderboard is worth more than a better-looking one that does not.** The Task 1 proxy predicted model changes closely (for example +0.0035 on the LB against +0.0037 locally). Its one large miss was v7, overstated about five times. The self-supervised step was invisible locally but gained +0.00138 twice on the LB. The Task 2 proxy was about 0.13 too high and often ranked candidates in the wrong order. For Task 2 we kept a rule: a change needs a reason to transfer to a month with different scenario seeds, in addition to a local gain.
- **Limits.**
  - Task 2 onset windows remain the weakest part.
  - Several larger Task 2 changes (split climatology retrains, month-ahead recalibration, adaptive gates) were neutral or negative on the leaderboard. Section 4 gives the numbers.
  - The private month may react differently from the public month to the climatology gate. Both finals use it, so this risk is not hedged; the LightGBM-only i5 would have been the hedge, at 0.0036 lower public score.
  - Task 1 reproduces at score level, not byte level, across operating systems.

## References


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- Van Zuylen, H. J., Willumsen, L. G. (1980). The most likely trip matrix estimated from traffic counts. *Transportation Research Part B* 14(3).
- Cascetta, E. (1984). Estimation of trip matrices from traffic counts and survey data: a generalized least squares estimator. *Transportation Research Part B* 18(4–5).


## Appendix A. Submission ledger

All public leaderboard submissions in order. Each row gives its three input files and its public score. "Reference" is the submission the change was built on, the same one the tables in Sections 3.3 and 4.6 compare against. It always differs from the row in exactly one task file (where no experiment base is stated, it is the best-scoring earlier such submission). Δ is the score difference against it. Divide Δ by 0.30 for the queue file and by 0.20 for the ODME file to get the change in that task's score. For the state file, Δ is the change in 0.35·S_state + 0.15·S_physics. Rows marked `diag_` are diagnostic files, not candidates (Section 2.2). Row 56 is the reproduction check of Section 6.2: final A rebuilt from the code package on Linux. File names match the candidate folders in the code history.

| # | Time (KST) | Submission | State | Queue | ODME | Public LB | Reference | Δ |
|---|---|---|---|---|---|---|---|---|
| 1 | 10-25 13:18 | b0_baseline | baseline | baseline | baseline | 0.55343 | | |

| # | Time (KST) | Submission | State | Queue | ODME | Public LB | Reference | Δ |
|---|---|---|---|---|---|---|---|---|
| 2 | 10-25 13:19 | q1_lgb_v1 | baseline | lgb_v1 | baseline | 0.69083 | b0_baseline | +0.13740 |
| 3 | 10-25 13:40 | o1_nnls_own | baseline | baseline | nnls_own | 0.58501 | b0_baseline | +0.03158 |
| 4 | 10-25 13:42 | o2_nnls_w75 | baseline | baseline | nnls_w75 | 0.56381 | o1_nnls_own | -0.02120 |
| 5 | 10-25 14:24 | q2_diag_onset | baseline | diag_v1_onset | baseline | 0.66649 | q1_lgb_v1 | -0.02434 |
| 6 | 10-26 09:33 | c1_gbm4c_lgb1_nnls | gbm_v4c | lgb_v1 | nnls_own | 0.86066 | | |
| 7 | 10-26 09:35 | c8_gbm8s2_lgb1_nnls | gbm_v8s2 | lgb_v1 | nnls_own | 0.86414 | c1_gbm4c_lgb1_nnls | +0.00348 |
| 8 | 10-26 09:36 | c3_gbm4c_notod35_nnls | gbm_v4c | lgb_notod | nnls_own | 0.85836 | c1_gbm4c_lgb1_nnls | -0.00230 |
| 9 | 10-26 09:37 | c4_gbm4c_robust_nnls | gbm_v4c | lgb_robust | nnls_own | 0.85454 | c1_gbm4c_lgb1_nnls | -0.00612 |
| 10 | 10-26 10:11 | c9_diag_persT | gbm_v4c | diag_v1_onset_persT | nnls_own | 0.84604 | c1_gbm4c_lgb1_nnls | -0.01462 |
| 11 | 10-27 09:04 | d1_diag_sensor | gbm_v4c | diag_sensor_split | nnls_own | 0.78677 | c1_gbm4c_lgb1_nnls | -0.07389 |
| 12 | 10-27 09:05 | d3_diag_famA | gbm_v4c | diag_fam_A | nnls_own | 0.77700 | c1_gbm4c_lgb1_nnls | -0.08366 |
| 13 | 10-27 09:06 | d4_diag_famB | gbm_v4c | diag_fam_B | nnls_own | 0.79194 | c1_gbm4c_lgb1_nnls | -0.06872 |
| 14 | 10-27 09:07 | d8_gbm8s2ss_lgb1_nnls | gbm_v8s2_ss | lgb_v1 | nnls_own | 0.86552 | c8_gbm8s2_lgb1_nnls | +0.00138 |
| 15 | 10-27 09:09 | d7_gbm8s2_lgb2_nnls | gbm_v8s2 | lgb_v2 | nnls_own | 0.86067 | c8_gbm8s2_lgb1_nnls | -0.00347 |
| 16 | 10-28 09:03 | d9_diag_zeros | gbm_v4c | diag_zeros | nnls_own | 0.63274 | c1_gbm4c_lgb1_nnls | -0.22792 |
| 17 | 10-28 09:05 | d5_diag_famC | gbm_v4c | diag_fam_C | nnls_own | 0.78545 | c1_gbm4c_lgb1_nnls | -0.07521 |
| 18 | 10-28 09:06 | d10_diag_onsetdil | gbm_v4c | diag_v1_onset_dil | nnls_own | 0.81607 | c1_gbm4c_lgb1_nnls | -0.04459 |
| 19 | 10-28 09:07 | e1_x3gfix_lgb1_nnls | gbm_v8s2x3g_fixed | lgb_v1 | nnls_own | 0.86478 | c8_gbm8s2_lgb1_nnls | +0.00064 |
| 20 | 10-28 09:09 | e2_x3gssfix_lgb1_nnls | gbm_v8s2x3g_ss_fixed | lgb_v1 | nnls_own | 0.86616 | e1_x3gfix_lgb1_nnls | +0.00138 |
| 21 | 10-29 09:22 | f1_x3gss3f_lgb1_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1 | nnls_own | 0.86631 | e2_x3gssfix_lgb1_nnls | +0.00015 |
| 22 | 10-29 12:02 | f2_ss3f_br03_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_br03 | nnls_own | 0.86559 | f1_x3gss3f_lgb1_nnls | -0.00072 |
| 23 | 10-29 14:18 | f3_ss3f_todsnap_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_todsnap | nnls_own | 0.86381 | f1_x3gss3f_lgb1_nnls | -0.00250 |
| 24 | 10-29 14:20 | f3_ss3f_ongphys_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongphys | nnls_own | 0.87189 | f1_x3gss3f_lgb1_nnls | +0.00558 |
| 25 | 10-29 14:25 | f4_ss3f_ongphys03_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongphys03 | nnls_own | 0.87240 | f3_ss3f_ongphys_nnls | +0.00051 |
| 26 | 10-30 09:06 | g1_ss3f_ongphys0_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongphys0 | nnls_own | 0.87003 | f4_ss3f_ongphys03_nnls | -0.00237 |
| 27 | 10-30 09:07 | g2_ss3f_onaug_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_onaug_ongphys03 | nnls_own | 0.86765 | f4_ss3f_ongphys03_nnls | -0.00475 |
| 28 | 10-30 09:09 | g3_ss3f_ongphysaug_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongphysaug | nnls_own | 0.87419 | f4_ss3f_ongphys03_nnls | +0.00179 |
| 29 | 10-30 09:19 | g4_ss3f_ongpa44_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongpa44 | nnls_own | 0.87505 | g3_ss3f_ongphysaug_nnls | +0.00086 |
| 30 | 10-30 09:25 | g5_ss3f_ongpa26_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongpa26 | nnls_own | 0.87518 | g4_ss3f_ongpa44_nnls | +0.00013 |

| # | Time (KST) | Submission | State | Queue | ODME | Public LB | Reference | Δ |
|---|---|---|---|---|---|---|---|---|
| 31 | 10-31 10:05 | h1_ss3f_ongpm26_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongpm26 | nnls_own | 0.87514 | g5_ss3f_ongpa26_nnls | -0.00004 |
| 32 | 10-31 10:07 | h2_ss3f_sq31_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongpa26_sq31 | nnls_own | 0.87701 | g5_ss3f_ongpa26_nnls | +0.00183 |
| 33 | 10-31 10:08 | h3_ss3f_ongpo26_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_v1_ongpo26 | nnls_own | 0.87622 | g5_ss3f_ongpa26_nnls | +0.00104 |
| 34 | 10-31 10:10 | h4_ss3f_onthr_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_onthr_g5 | nnls_own | 0.87738 | g5_ss3f_ongpa26_nnls | +0.00220 |
| 35 | 10-31 10:16 | h5_ss3f_onthr_sq31_nnls | gbm_v8s2x3g_ss3f_fixed | lgb_onthr_sq31 | nnls_own | 0.87922 | g5_ss3f_ongpa26_nnls | +0.00404 |
| 36 | 11-01 09:03 | i1_v7_onthr_sq31_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_onthr_sq31 | nnls_own | 0.87951 | h5_ss3f_onthr_sq31_nnls | +0.00029 |
| 37 | 11-01 09:05 | i2_v7_pao33_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_onthr_sq31_pao33 | nnls_own | 0.87943 | i1_v7_onthr_sq31_nnls | -0.00008 |
| 38 | 11-01 09:07 | i3_v7_athr_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_onthr_sq31_athr | nnls_own | 0.87906 | i1_v7_onthr_sq31_nnls | -0.00045 |
| 39 | 11-01 09:09 | i4_v7_hmix_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_onthr_sq31_hmix | nnls_own | 0.88006 | i1_v7_onthr_sq31_nnls | +0.00055 |
| 40 | 11-01 09:11 | i5_v7_pao1535_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_onthr_sq31_pao1535 | nnls_own | 0.88019 | i1_v7_onthr_sq31_nnls | +0.00068 |
| 41 | 11-02 09:03 | k_v7_cpu3go50_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3_go50 | nnls_own | 0.87876 | i5_v7_pao1535_nnls | -0.00143 |
| 42 | 11-02 09:04 | j1_rf7_pao1535_nnls | gbm_v8s2rx3g_ss3f_ramp_rf7 | lgb_onthr_sq31_pao1535 | nnls_own | 0.88019 | i5_v7_pao1535_nnls | +0.00000 |
| 43 | 11-02 09:06 | k_v7_cpu3go70_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3_go70 | nnls_own | 0.87778 | i5_v7_pao1535_nnls | -0.00241 |
| 44 | 11-02 09:07 | k_v7_cpu3go30_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3_go30 | nnls_own | 0.87905 | i5_v7_pao1535_nnls | -0.00114 |
| 45 | 11-02 09:08 | j2_v7_pao1535hmix_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_pao1535_hmix | nnls_own | 0.88013 | i5_v7_pao1535_nnls | -0.00006 |
| 46 | 11-03 09:03 | n_v7_scr75_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_i5_scr75 | nnls_own | 0.87527 | i5_v7_pao1535_nnls | -0.00492 |
| 47 | 11-03 09:04 | n_v7_sca75_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_i5_sca75 | nnls_own | 0.87366 | i5_v7_pao1535_nnls | -0.00653 |
| 48 | 11-03 09:05 | n_v7_sca5_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_i5_sca5 | nnls_own | 0.87285 | i5_v7_pao1535_nnls | -0.00734 |
| 49 | 11-03 09:07 | m_v7_cpu3ggo50_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3g_go50 | nnls_own | 0.88376 | i5_v7_pao1535_nnls | +0.00357 |
| 50 | 11-03 09:07 | m1_v7_sclim75p_nnls | gbm_v8s2rx3g_ss3f_ramp | lgb_i5_sclim75p | nnls_own | 0.88019 | i5_v7_pao1535_nnls | +0.00000 |
| 51 | 11-04 09:03 | p_v7_cpu3ggo70_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3g_go70 | nnls_own | 0.88324 | m_v7_cpu3ggo50_nnls | -0.00052 |
| 52 | 11-04 09:04 | p_v7_cpu3gc00_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3g_go50_c00 | nnls_own | 0.88215 | m_v7_cpu3ggo50_nnls | -0.00161 |
| 53 | 11-04 09:06 | m_v7_cpu3ggo30_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3g_go30 | nnls_own | 0.88263 | m_v7_cpu3ggo50_nnls | -0.00113 |
| 54 | 11-04 09:07 | p_v7_cpu3gc07_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3g_go50_c07 | nnls_own | 0.88055 | m_v7_cpu3ggo50_nnls | -0.00321 |
| 55 | 11-04 09:09 | q_v7_cpu3gago50_nnls | gbm_v8s2rx3g_ss3f_ramp | cpu3ga_go50 | nnls_own | 0.88315 | m_v7_cpu3ggo50_nnls | -0.00061 |
| 56 | 11-05 17:00 | r_pkg_gb10_A | v7, package rebuild | cpu3g_go50, package rebuild | nnls_own, package rebuild | 0.88376 | m_v7_cpu3ggo50_nnls | +0.00000 |